
FLAVOURBENCH: Epicure as an Executable Oracle for Ranking Culinary Agents

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Abstract

Culinary reasoning is usually evaluated with broad preference judgments or static question banks. Both make it difficult to separate remembered food knowledge from the ability to query a domain system and use its result. We introduce FLAVOURBENCH, an automated benchmark built from the versioned EPICURE ingredient runtime. EPICURE deterministically compiles four families of multiple-choice tasks—nearest-neighbour substitution, exact pairing affinity, flavour-axis comparison, and cuisine projection—together with their answer keys. No language model writes or judges a task.

We evaluate 20 frontier models on the same 32 tasks in two matched conditions: model alone and model with read-only EPICURE access. The release contains 640 paired trials and 1280 scored arms. GPT-5.6 Sol Pro ranks first with an Epicure Benchmark Score of 62.5, while Nemotron 3 Ultra scores 0.0. Across the panel, tool access changes exact-choice accuracy from 43.9% to 93.9% (+50.0 percentage points). The paired design therefore provides both a conventional model ranking and a direct estimate of domain-tool uplift.

The benchmark is fully replayable from a single content-addressed release containing the task compiler outputs, answer keys, routes, final responses, and projected tool traces. Beyond evaluation, the same exact reward can support rejection sampling, preference construction, and reinforcement learning. FLAVOURBENCH turns a culinary knowledge system into an executable oracle: it measures what a model knows, what it can retrieve, and whether tool access actually improves the answer.

1 Introduction

Benchmarks are easiest to trust when their ground truth is external to the models being tested. Code can be executed, database states can be inspected, and games can be scored. Culinary reasoning rarely has that luxury. Ingredient substitutions, pairings, technique choices, and cuisine associations are commonly assessed by another language model or by a small set of human preferences. The evaluator then becomes part of the construct.

FLAVOURBENCH takes a different route. It starts from EPICURE, a read-only culinary runtime with 1,790 ingredients represented in a 300-dimensional space [1, 2]. The benchmark calls four deterministic operations, converts their results into four-choice questions, and records the exact answer before any evaluated model is contacted. The resulting task is compact enough to rerun whenever the frontier changes, yet structured enough to expose different kinds of culinary reasoning.

The experimental design asks two questions at once. First, how accurately can a model recover EPICURE’s answers without access to the runtime? This is the public model ranking. Second, what changes when the same model receives the exact read-only operation needed to solve the task? This

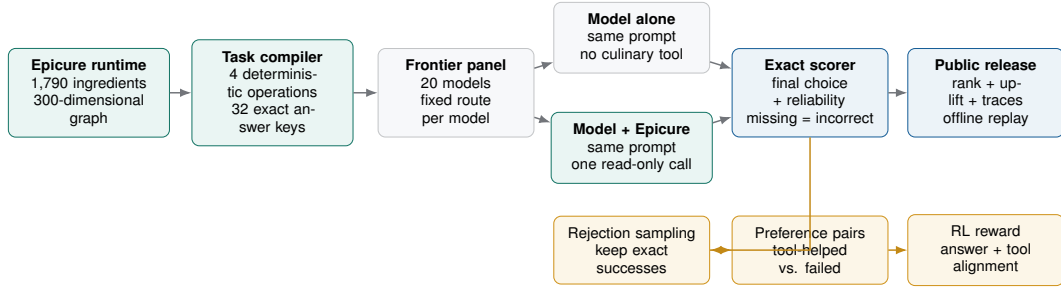


Figure 1: FLAVOURBENCH turns EPICURE into an executable oracle. The upper path is the released benchmark: deterministic task construction, matched model calls, exact scoring, and offline replay. The lower path reuses the same score as training data without introducing an LLM judge.

is the paired tool-uplift experiment. Every model sees every task in both conditions, so differences are not confounded by task allocation.

Our release evaluates 20 current endpoints, including GPT-5.6 Sol, Terra, and Luna; Claude Fable, Opus, and Sonnet 5; Gemini 3.1 Pro and 3.6 Flash; Grok 4.5; Kimi K3; Qwen 3.8 Max; GLM 5.2; two DeepSeek V4 variants; MiniMax M3; Nemotron 3 Ultra; Mistral Large 2512; Tencent HY 3; and Cohere Command A Plus and Command A Reasoning. Kimi and both Cohere models use direct provider routes. Every other endpoint uses an exact, no-fallback OpenRouter route selected before the run.

The paper makes four contributions:

1. an automated culinary benchmark whose questions and answer keys are generated by a versioned executable oracle rather than a model judge;
2. a common-task frontier evaluation with 640 matched pairs, exact missing-response penalties, and a public 20-row leaderboard;
3. a matched tool-access comparison that reports both aggregate uplift and model-family interactions; and
4. a training interface that converts the same deterministic reward into rejection sampling, pairwise preference data, or reinforcement learning signals.

The headline outcome is concrete: EPICURE is used twice, first to create the benchmark and then as the intervention being measured. That makes the same culinary system both an evaluation oracle and a measurable tool intervention.

2 Epicure-native ground truth

2.1 A versioned executable oracle

The evaluated target is not a chef’s universal preference. It is the output of a specific, versioned culinary computation. The release binds an EPICURE application hash, evidence-bundle hash, release identifier, ingredient count, and embedding dimensionality. Rebuilding the task set against a different runtime necessarily produces a different task-set hash.

This definition is narrow by design. It makes every answer independently recomputable and allows models to be compared without asking another model which response sounds better. It also gives the score a direct interpretation: accuracy is agreement with the executable culinary model under the published task construction.

2.2 Four balanced task families

The compiler creates eight tasks in each of four families. Seeds and ingredient pairs are fixed before evaluation. For every task, EPICURE is called once to obtain the reference result. The correct answer is extracted programmatically, three distractors are constructed without model generation, and the four options are shuffled with a task-ID-derived seed.

Family	Epicure operation	Exact target	Example query	Tasks
Substitution	<code>neighbors</code>	first ranked neighbour	top neighbour for tomato	8
Composition	<code>pairing_score</code>	four-decimal affinity	tomato + basil	8
Cookability	<code>compare_on_axis</code>	larger flavour projection	coffee vs. chocolate on bitter	8
Evidence	<code>cultural_profile</code>	highest cuisine cosine	cuisine direction for kimchi	8

Table 1: The task set is balanced by construction. Each family maps one deterministic EPICURE operation to a four-choice answer. Chance accuracy is 25%.

The task families deliberately mix retrieval and computation. Neighbour and cuisine questions test whether a model knows relationships encoded by the graph. Pairing questions require an exact numeric result that is difficult to guess. Axis questions test a comparative claim rather than a memorised scalar. Together they prevent the leaderboard from reducing to one operation or one kind of culinary trivia.

2.3 Prompt and answer contract

Each prompt names the relevant operation and gives four choices. Models may explain briefly, but the last answer line must be `FINAL_CHOICE: X`. The parser accepts exactly one of A–D on that marker line. A missing marker, abnormal provider finish, timeout, or absent response scores zero. The rule is intentionally simple enough to implement in a standalone verifier.

The complete task set contains no hidden human label and no model-authored item. The benchmark is therefore suitable for automatic reruns, CI-style regression tests, and training loops where thousands of candidate responses must be scored consistently.

3 Evaluation design

3.1 Matched tool-off and tool-on arms

For model m , task t , and condition $c \in \{0, 1\}$, let Y_{mtc} equal one when the response finishes normally and its final choice matches the EPICURE answer key. Condition $c = 0$ removes the tool catalog; condition $c = 1$ exposes a minimal read-only EPICURE call contract. The prompt, route, temperature, top- p , seed request, output limit, and answer parser are otherwise identical.

Each tool-enabled arm may make one successful culinary call. The runner records the tool name, arguments, latency, error flag, and a hash of the returned payload. Tool output is bounded before it is appended to model context. Automatic provider fallback is disabled: a route failure remains a failure for that route.

The model-alone benchmark score is

$$S_m = 100 \frac{1}{4} \sum_{f=1}^4 \frac{1}{8} \sum_{t \in f} Y_{mt0}. \quad (1)$$

Because every family contains eight tasks, this equals ordinary accuracy over 32 tasks while making the intended family weighting explicit. The public rank sorts by S_m , then by tool-enabled accuracy, tool-enabled parse reliability, and model ID.

Tool uplift is the matched difference

$$U_m = 100 \frac{1}{32} \sum_t (Y_{mt1} - Y_{mt0}), \quad (2)$$

reported in percentage points. We also retain the four-cell paired decomposition: both correct, model-alone only, EPICURE only, and neither correct. This is more informative than two unrelated accuracy bars because it shows where the tool actually changes an answer.

3.2 Reliability and uncertainty

Reliability is the fraction of assigned arms that end normally with a parseable final marker. Missing arms are never dropped from the denominator. For each 32-task accuracy we report a 95% Wilson

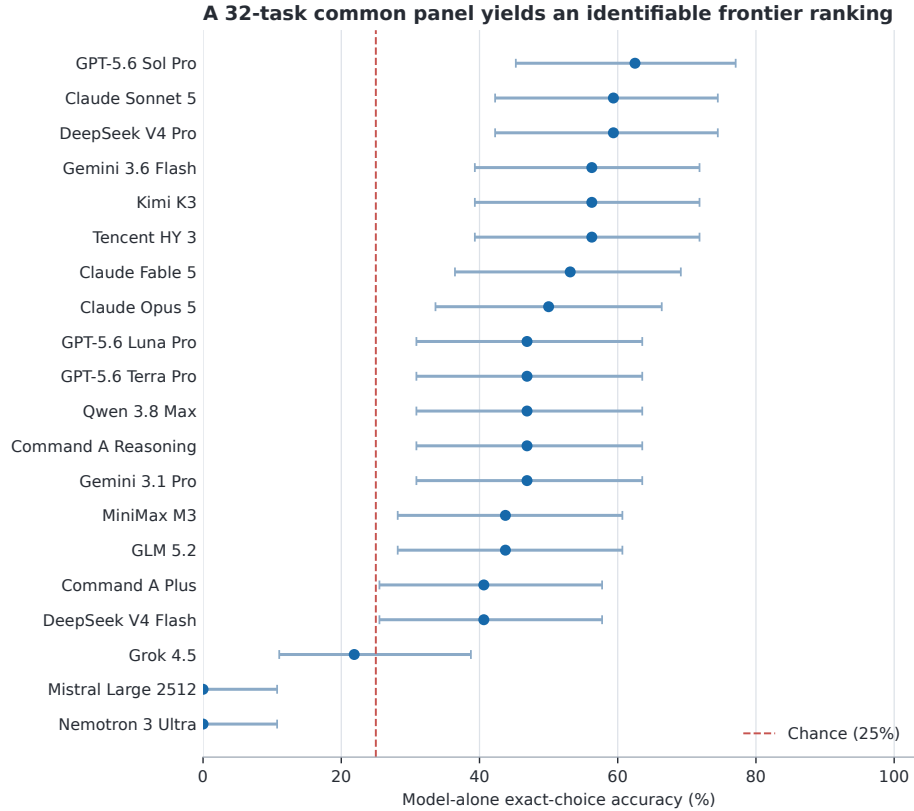


Figure 2: Model-alone accuracy against EPICURE answer keys. Points are exact accuracy over the common 32-task panel; bars are 95% Wilson intervals. The dashed line is four-choice chance.

interval. These intervals describe sampling uncertainty over the published task panel; they do not alter rank order.

The benchmark is designed for visible differences, not hundredths of a point. One answer changes the score by 3.125 points. Family breakdowns and the full pair matrix are published so tied or nearby aggregate scores can be inspected rather than over-interpreted.

3.3 Twenty-model frontier panel

The panel was fixed before the final calls. It includes three GPT-5.6 tiers rather than treating one deployment as the entire OpenAI family; three Claude 5 tiers; two Gemini tiers; two DeepSeek V4 routes; both current Cohere Command A routes; and a broad set of current independent and open-weight families. Kimi K3 runs through Kimi’s Anthropic-compatible endpoint. Command A Plus and Command A Reasoning run through Cohere V2 Chat. Qwen 3.8 Max and GLM 5.2 use explicit OpenRouter routes in the automated batch [5–7, 11, 14, 15]. The exact 20-row route table appears in Appendix C.

4 Results

4.1 The Epicure Benchmark leaderboard

GPT-5.6 Sol Pro ranks first with 20/32 correct and a score of 62.5. Nemotron 3 Ultra ranks twentieth with 0/32 and 0.0. The gap is large enough to be visible under both the point estimates and the task-level matrix. The complete ordering is shown in Table 2; Figure 2 adds interval estimates without replacing the exact rank.

Rank	Model	Base	+Epicure	Δ	Base OK	Tool OK
1	GPT-5.6 Sol Pro	62.5	100.0	+37.5	96.9	100.0
2	Claude Sonnet 5	59.4	100.0	+40.6	100.0	100.0
3	DeepSeek V4 Pro	59.4	93.8	+34.4	100.0	93.8
4	Gemini 3.6 Flash	56.2	100.0	+43.8	100.0	100.0
5	Kimi K3	56.2	100.0	+43.8	100.0	100.0
6	Tencent HY 3	56.2	100.0	+43.8	100.0	100.0
7	Claude Fable 5	53.1	100.0	+46.9	93.8	100.0
8	Claude Opus 5	50.0	100.0	+50.0	100.0	100.0
9	GPT-5.6 Luna Pro	46.9	100.0	+53.1	100.0	100.0
10	GPT-5.6 Terra Pro	46.9	100.0	+53.1	100.0	100.0
11	Qwen 3.8 Max	46.9	100.0	+53.1	87.5	100.0
12	Command A Reasoning	46.9	96.9	+50.0	93.8	100.0
13	Gemini 3.1 Pro	46.9	96.9	+50.0	93.8	96.9
14	MiniMax M3	43.8	100.0	+56.2	96.9	100.0
15	GLM 5.2	43.8	96.9	+53.1	100.0	96.9
16	Command A Plus	40.6	100.0	+59.4	100.0	100.0
17	DeepSeek V4 Flash	40.6	100.0	+59.4	75.0	100.0
18	Grok 4.5	21.9	100.0	+78.1	31.2	100.0
19	Mistral Large 2512	0.0	93.8	+93.8	0.0	100.0
20	Nemotron 3 Ultra	0.0	0.0	+0.0	0.0	0.0

Table 2: The public 20-model leaderboard. “Base” is model-alone exact-choice accuracy; “+Epicure” is the matched tool-enabled accuracy; Δ is the percentage-point change; “Base OK” and “Tool OK” are parseable normal-completion rates in the two arms. Rank is determined by the model-alone Epicure Benchmark Score, with the published tie-breaks.

Family	Epicure operation	Tasks	Base	+Epicure
Substitution	<code>neighbors</code>	8	18.1	94.4
Composition	<code>pairing_score</code>	8	40.6	94.4
Cookability	<code>compare_on_axis</code>	8	62.5	92.5
Evidence	<code>cultural_profile</code>	8	54.4	94.4

Table 3: Panel-average performance by family and the corresponding reference operation.

This is an automated model ranking, not a human preference table. The answer key exists before the model call, every model receives the same questions, and all missing outputs remain incorrect. The score can therefore be rerun as the roster changes without recruiting judges or prompting an LLM evaluator.

The reliability columns separate culinary mistakes from execution failures. At the reliability floor, Mistral Large 2512 and Nemotron 3 Ultra emit a valid final marker on 0.0% of their arms. This is part of endpoint performance rather than discarded noise: a response that never commits to A–D cannot solve an exact-choice task.

4.2 What Epicure changes

Across all 640 model–task pairs, accuracy moves from 43.9% without the tool to 93.9% with it, an aggregate change of +50.0 points. The average conceals substantial heterogeneity: 19 models improve, 0 decline, and 1 are unchanged. Mistral Large 2512 has the largest increase (+93.8 points); Nemotron 3 Ultra has the smallest change (+0.0 points).

Tool access is not guaranteed to help. A model can select the wrong operation, copy the wrong field, or abandon a correct prior belief after seeing an irrelevant result. Conversely, a model with weak memorised culinary knowledge can recover the exact target when it chooses the reference operation. The paired design exposes both behaviours.

4.3 Families explain aggregate ties

The four task families have different demands. Pairing questions require an exact four-decimal value; substitution and cultural questions can sometimes be answered from broad food knowledge; axis comparisons require a directional decision in EPICURE’s latent space. Figure 4 shows that similar total scores can arise from different profiles.

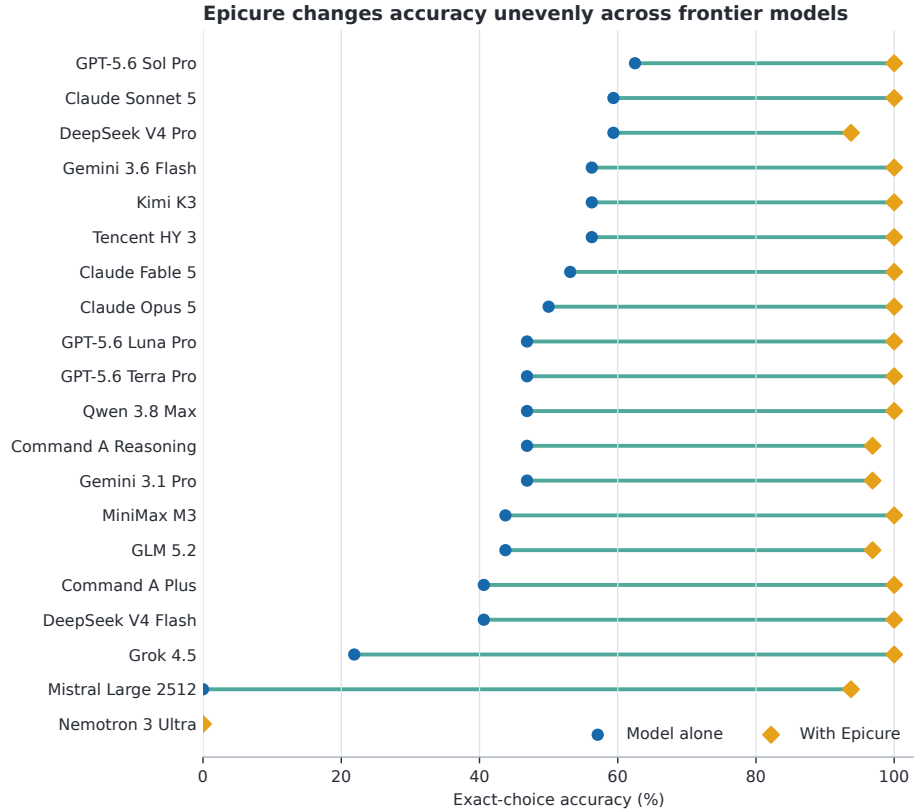


Figure 3: Matched model-alone and tool-enabled accuracy. The line segment for each model is an intervention contrast over the same 32 tasks, not a comparison between different samples.

146 4.4 Every model-task pair is inspectable

147 Aggregate bars can hide a benchmark bug or a single pathological task. Figure 5 therefore publishes
 148 all 640 paired outcomes. Blue cells are stable knowledge: both conditions are correct. Gold cells are
 149 direct tool wins. Red cells identify regressions after tool access. Grey cells show questions neither
 150 condition solved.

151 4.5 Cohere, Kimi, Qwen, and route fidelity

152 Both Cohere models are first-class members of the public panel, not appendix-only smokes. Command
 153 A Plus is ranked 16 at 40.6 and changes by +59.4 points with EPICURE; Command A Reasoning is
 154 ranked 12 at 46.9 and changes by +50.0 points. Kimi K3 is rank 5 through the direct Kimi endpoint.
 155 Qwen 3.8 Max is rank 11 through an explicit pay-as-you-go OpenRouter route, and GLM 5.2 is rank
 156 15. No route silently changes during generation.

157 The two direct Cohere routes each produced all 64 assigned arms. Their returned identifiers are `comm`
 158 `and-a-plus-05-2026` and `command-a-reasoning-08-2025`, respectively; the direct Kimi route
 159 likewise produced all 64 arms under returned identifier `k3`. The offline verifier checks these provider,
 160 identity, and arm-count contracts before accepting the release.

161 This matters because a model name is not a transport contract. The release stores the requested model,
 162 returned model, provider, route, generation ID, latency, token counts, and response hash. Direct and
 163 routed endpoints can therefore be analysed separately without pretending they are interchangeable
 164 deployments.

165 4.6 Accuracy, latency, and uplift are different axes

166 Figure 6 plots model-alone score against median arm latency. Marker colour encodes uplift. A fast
167 model can be inaccurate; a high-scoring model can receive little benefit from the tool; a lower-ranked
168 model can be the best candidate for tool augmentation. These are separate deployment decisions, and
169 the release preserves all three.

170 5 From benchmark score to training reward

171 The same environment can supervise more than a leaderboard. For each generated trajectory τ , define
172 four deterministic components:

$$r_{\text{answer}}(\tau) = \mathbb{1}[\hat{y}(\tau) = y_t], \quad (3)$$

$$r_{\text{format}}(\tau) = \mathbb{1}[\text{normal parseable final}], \quad (4)$$

$$r_{\text{call}}(\tau) = \mathbb{1}[\text{successful Epicure call}], \quad (5)$$

$$r_{\text{align}}(\tau) = \mathbb{1}[\text{reference call matches}]. \quad (6)$$

173 The public leaderboard uses r_{answer} in the model-alone condition. A tool-agent reward can use

$$r(\tau) = r_{\text{answer}} r_{\text{format}} + \lambda r_{\text{answer}} r_{\text{call}} + \mu r_{\text{align}}, \quad (7)$$

174 with λ and μ chosen for the training objective. The answer term prevents reward hacking through
175 irrelevant tool calls; the alignment term distinguishes reaching the right answer by the intended
176 evidence path from accidental success.

177 Three training regimes follow naturally. **Rejection sampling** generates multiple responses and retains
178 exact successes. **Preference optimisation** pairs a successful trajectory with a failed response to
179 the same task and model checkpoint. **Reinforcement learning** queries the runtime online and uses
180 Equation (7) as a verifiable reward. New ingredient seeds create fresh tasks, which reduces direct
181 memorisation of the published 32-item panel.

182 This paper releases the executable reward interface and the paired evidence that motivates it; training
183 a new checkpoint is the next experiment. That experiment should use disjoint generated seeds and
184 keep the public leaderboard as a final evaluation split.

185 6 Reproducibility and release

186 The public release is one JSON document with hash 603e4aa4db86... It contains the 20 route
187 records, 32 full prompts, answer keys and reference EPICURE results, 1,280 observation rows, final
188 answer markers, latency and cost fields, and bounded tool-trace projections. 1195 provider responses
189 were observed; 85 missing arms remain explicit zero-scored rows.

190 The leaderboard can be reconstructed without a network connection, provider key, GPU, or running
191 EPICURE service:

```
192 python3 -I reproduce_epicure_native.py \  
193 --release generated/epicure-native/\  
194 epicure-native-release.json
```

195 The verifier rejects duplicate JSON keys, non-finite numbers, a changed content address, an incom-
196 plete $20 \times 32 \times 2$ grid, or any leaderboard value that differs from recomputation. The paper tables
197 and figures are generated from the same release, so a changed score cannot leave a stale chart behind.

198 Reproducing the original hosted calls is a different operation from replaying the result. The release
199 records exact routes and request policies, but provider deployments can change. A fresh frontier run
200 should generate a new manifest and release rather than overwrite this snapshot.

201 7 Discussion

202 **What the rank means.** The Epicure Benchmark Score measures how often a model can recover
203 answers defined by a published culinary runtime. It is not a universal measure of cooking skill. Its

Benchmark	Scope	Per-answer target	Deterministic score	Matched tool contrast
Chatbot Arena	general chat	pairwise human preference	No	No
WildBench	general chat	checklist-guided model judgment	No	No
BFCL	function calling	structured call correctness	Yes	No
τ -bench	interactive agents	environment goal state	Yes	No
FLAVOURBENCH	culinary agents	versioned EPICURE operation	Yes	Yes

Table 4: FLAVOURBENCH complements broad preference and tool-use benchmarks. Its distinctive unit is the same model–task pair measured without and with the domain oracle that generated the answer.

value is that the target is clear, model-independent, and executable. The four families make the ranking broader than one API call while keeping every item auditable.

Do we need humans? Not for this leaderboard. Humans are useful when the target is preference, creativity, safety, or the quality of an open-ended recipe. They are unnecessary when the question is whether a model’s choice matches a deterministic runtime result. FLAVOURBENCH uses that distinction to automate the core ranking while leaving subjective culinary evaluation as a separate track.

Why pair tool access? A standalone benchmark says which model best predicts EPICURE. The paired intervention says whether EPICURE itself helps each model. That second result is operationally important: a strong base model may ignore tools, while a weaker model may become competitive when grounded. The dumbbell and pair matrix expose this directly.

Scaling the benchmark. The current task compiler can generate more items by expanding seed lists while preserving the same four operations. A larger release should reserve unseen ingredients and pairs, deduplicate near-equivalent questions, and measure stability across runtime versions. The public 32-task panel is best viewed as a frontier snapshot and a reproducible template for larger hidden sets.

8 Related work

HELM argues for explicit scenarios and metrics, while recent benchmark audits emphasise validity, contamination, and transparent reporting [12, 19]. Chatbot Arena and WildBench evaluate broad answer quality through preferences or checklists [3, 13]. FLAVOURBENCH instead targets an executable domain model and needs no judge for its primary score.

ToolLLM, BFCL, and τ -bench make tool invocation or environment state observable [16, 17, 24]. FLAVOURBENCH adds a matched no-tool condition, so tool access becomes an intervention rather than a requirement shared by every arm. LiveBench, LiveCodeBench, GPQA, MMLU-Pro, and SWE-bench illustrate complementary strategies for reducing contamination or grounding answers in difficult tasks and executable outcomes [9, 10, 18, 21, 22].

Culinary datasets such as FlavorDB and Recipe1M organise ingredient and recipe evidence [8, 20]. CookingSense and planning-oriented culinary work test commonsense and procedural reasoning [4, 23]. EPICURE occupies a different layer: it exposes a callable structured representation whose outputs can be compiled into exact evaluation targets.

9 Limitations

The published panel has 32 tasks and therefore resolves large score differences better than close ones. The ingredient seeds are English identifiers and cover only four EPICURE operations. Hosted model

237 routes are a dated snapshot, and identical public prompts can eventually be memorised. Finally,
238 agreement with EPICURE inherits the runtime’s representation choices; it should not be read as a
239 food-safety or cultural-authenticity certification.

240 These limits have direct remedies: generate held-out seeds, add runtime operations, report multiple
241 versioned panels, and keep subjective recipe-quality evaluation separate. None requires replacing the
242 exact automated score with an opaque judge.

243 **10 Conclusion**

244 FLAVOURBENCH supplies a public automated leaderboard for culinary agents. EPICURE creates the
245 questions, fixes the answers, and serves as the paired tool intervention. On the 20-model common
246 panel, GPT-5.6 Sol Pro ranks first at 62.5 and Nemotron 3 Ultra ranks twentieth at 0.0. Tool access
247 changes panel accuracy by +50.0 points, with strongly model-dependent effects. Every number is
248 recoverable from the released 1,280-arm grid.

249 The broader result is architectural: a domain model can be more than a retrieval plugin. When its
250 outputs are deterministic and versioned, it becomes an oracle for evaluation and a reward source for
251 learning. FLAVOURBENCH demonstrates that loop end to end.

252 A Real prompts, answers, and tool calls

253 A benchmark paper should show what was actually asked. The four cases below are selected
254 mechanically, one per family. Where possible, the builder chooses a model–task pair that is wrong
255 without EPICURE and correct with it. Each case includes the full prompt, both final answers, the
256 actual operation and arguments, and the exact answer key.

257 **Case 1: Substitution with GPT-5.6 Sol Pro. Prompt.** FlavourBench Epicure-native exact task.
258 Use the read-only culinary evidence tools if they are available; otherwise answer from your own
259 knowledge.

260 According to Epicure’s ‘neighbors’ result for ‘tomato’ with top_k=4, which ingredient is ranked first?

261 Choices: A. oregano B. bell_pepper C. olive_oil D. onion

262 In your final answer, explain briefly if useful and finish on its own line with exactly ‘FINAL_CHOICE:
263 X’, replacing X with A, B, C, or D.

264 **Model alone.** FINAL_CHOICE: D **With Epicure.** FINAL_CHOICE: B

265 **Epicure call.** neighbors({"ingredient": "tomato", "top_k": 4}). The answer key is
266 FINAL_CHOICE: B.

267 **Epicure result.** {"ingredient": "tomato", "neighbors": [{"name": "bell_pepper", "rank":
268 1, "sim": 0.4799}, {"name": "olive_oil", "rank": 2, "sim": 0.4786}, {"name": "oregano",
269 "rank": 3, "sim": 0.4646}, {"name": "onion", "rank": 4, "sim": 0.4567}]}

270 **Case 2: Composition with DeepSeek V4 Pro. Prompt.** FlavourBench Epicure-native exact task.
271 Use the read-only culinary evidence tools if they are available; otherwise answer from your own
272 knowledge.

273 What exact four-decimal ‘pairing_score’ does Epicure return for ‘tomato’ and ‘basil’?

274 Choices: A. 0.2707 B. 0.2407 C. 0.3207 D. 0.1907

275 In your final answer, explain briefly if useful and finish on its own line with exactly ‘FINAL_CHOICE:
276 X’, replacing X with A, B, C, or D.

277 **Model alone.** FINAL_CHOICE: B **With Epicure.** FINAL_CHOICE: A

278 **Epicure call.** pairing_score({"ingredient_a": "tomato", "ingredient_b":
279 "basil"}). The answer key is FINAL_CHOICE: A.

280 **Epicure result.** {"all_pairs_range": {"median": 0.092, "p10": 0.0197, "p25": 0.0521, "p75":
281 0.1392, "p90": 0.1899}, "pairing_score": 0.2707, "percentile_label": "very high (>=p90)",
282 "resolved_a": "tomato", "resolved_b": "basil"}

283 **Case 3: Cookability with GPT-5.6 Sol Pro. Prompt.** FlavourBench Epicure-native exact task.
284 Use the read-only culinary evidence tools if they are available; otherwise answer from your own
285 knowledge.

286 Which statement matches Epicure’s ‘compare_on_axis’ result for ‘coffee’ versus ‘chocolate’ on
287 ‘cf_bitter’?

288 Choices: A. coffee has the higher projection B. the projections are equal to four decimals C. Epicure
289 cannot resolve one or both ingredients D. chocolate has the higher projection

290 In your final answer, explain briefly if useful and finish on its own line with exactly ‘FINAL_CHOICE:
291 X’, replacing X with A, B, C, or D.

292 **Model alone.** FINAL_CHOICE: A **With Epicure.** FINAL_CHOICE: D

293 **Epicure call.** compare_on_axis({"axis": "cf_bitter", "ingredient_a": "coffee",
294 "ingredient_b": "chocolate"}). The answer key is FINAL_CHOICE: D.

295 **Epicure result.** {"axis": "cf_bitter", "axis_range": {"p10": -0.0877, "p90": 0.1429},
296 "delta": 0.0835, "label_a": "moderate", "label_b": "high", "projection_a": 0.0076,
297 "projection_b": 0.0911, "resolved_a": "coffee", "resolved_b": "chocolate"}

298 **Case 4: Evidence with Grok 4.5. Prompt.** FlavourBench Epicure-native exact task. Use the
299 read-only culinary evidence tools if they are available; otherwise answer from your own knowledge.
300 Which cuisine direction has the highest cosine score in Epicure’s ‘cultural_profile’ result for ‘miso’?
301 Choices: A. Japanese B. East_Asian C. Southeast_Asian D. South_Asian
302 In your final answer, explain briefly if useful and finish on its own line with exactly ‘FINAL_CHOICE:
303 X’, replacing X with A, B, C, or D.
304 **Model alone.** {"name": "cultural_profile", "arguments": {"ingredient":
305 "miso"}} **With Epicure.** FINAL_CHOICE: A
306 **Epicure call.** cultural_profile({"ingredient": "miso"}). The answer key is
307 FINAL_CHOICE: A.
308 **Epicure result.** {"cuisines": {"East_Asian": {"range": {"p10": -0.1649, "p90":
309 0.3039}, "score": 0.1625}, "Eastern_European": {"range": {"p10": -0.1466, "p90":
310 0.1034}, "score": -0.1494}, "Japanese": {"range": {"p10": -0.1359, "p90": 0.1896},
311 "score": 0.3227}, "Latin_American": {"range": {"p10": -0.1672, "p90": 0.1281},
312 "score": -0.1538}, "Mediterranean": {"range": {"p10": -0.2323, "p90": 0.206}, "score":
313 -0.1198}, "South_Asian": {"range": {"p10": -0.0808, "p90": 0.071}, "score": -0.0585},
314 "Southeast_Asian": {"range": {"p10": -0.1625, "p90": 0.193}, "score": 0.0426},
315 "Western_Atlantic": {"range": {"p10": -0.2304, "p90": 0.183}, "score": -0.1546}},
316 "note": "Cosine similarity to each cuisine direction. Use p10/p90 to judge whether a score
317 is typical or extreme.", "resolved": "miso"}
318 These examples make the construct visible. The model-alone arm tests knowledge or inference about
319 the hidden graph. The tool-enabled arm tests whether the model can identify the relevant operation,
320 form valid arguments, read the returned structure, and preserve the answer contract. A correct choice
321 with the wrong operation still earns answer credit, but the release separately reports whether the
322 reference operation was matched. Across the run there are 606 projected EPICURE calls and 606
323 model-task pairs matching the reference operation.

324 B Additional result figures

325 C Exact execution routes

Slot	Model	Execution route
1	GPT-5.6 Sol Pro	openrouter: openai/flex
6	Claude Sonnet 5	openrouter: amazon-bedrock/claude-on-aws
13	DeepSeek V4 Pro	openrouter: baidu/fp8
8	Gemini 3.6 Flash	openrouter: google-vertex/global/flex
10	Kimi K3	kimi_direct: kimi-code-direct
18	Tencent HY 3	openrouter: baidu/fp8
4	Claude Fable 5	openrouter: amazon-bedrock
5	Claude Opus 5	openrouter: amazon-bedrock/claude-on-aws
3	GPT-5.6 Luna Pro	openrouter: openai/flex
2	GPT-5.6 Terra Pro	openrouter: openai/flex
11	Qwen 3.8 Max	openrouter: alibaba
20	Command A Reasoning	cohere_direct: cohere-direct
7	Gemini 3.1 Pro	openrouter: google-vertex/global/flex
15	MiniMax M3	openrouter: together
12	GLM 5.2	openrouter: coreweave/fp4
19	Command A Plus	cohere_direct: cohere-direct
14	DeepSeek V4 Flash	openrouter: cloudflare/fp8
9	Grok 4.5	openrouter: xai/zdr
17	Mistral Large 2512	openrouter: mistral
16	Nemotron 3 Ultra	openrouter: deepinfra/fp4

Table 5: All 20 model routes in the released leaderboard. Routes are selected before generation; automatic fallback is disabled.

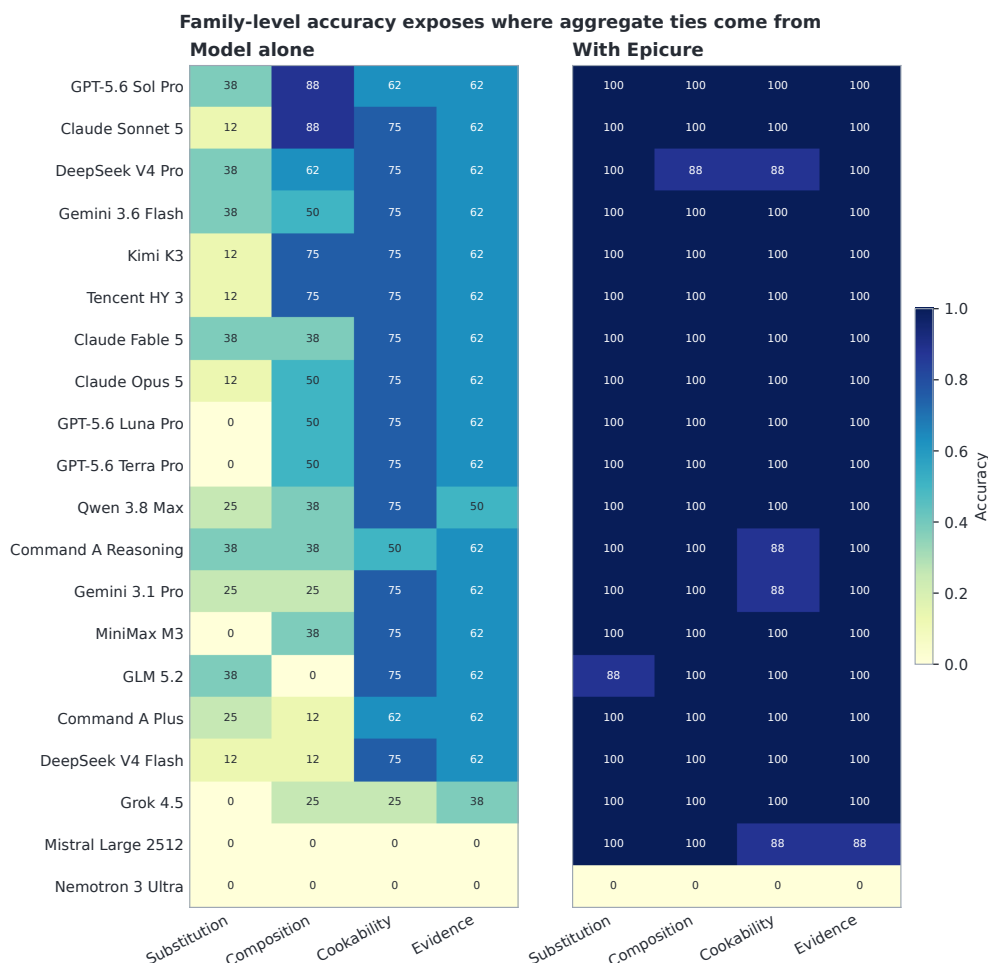


Figure 4: Accuracy by model, condition, and task family. Values are percentages over eight tasks. The paired heatmaps show where tool access repairs a knowledge gap and where it introduces one.

D Protocol details

All arms request temperature 0, top- p 1, and seed 20260810 where the provider supports it. The final response budget is 2,048 tokens. The tool-enabled arm has one tool round, one call total, a 4,096-byte result bound, and a 4,096-byte cumulative result bound. At most two provider attempts are allowed, and an ambiguously delivered request is never replayed. The final marker is parsed from plain text for transport portability; direct providers may use native structured generation internally and normalize to the same marker.

E Data fields

Each observation binds model ID, task ID, condition, source status, source and response hashes, returned model and provider, finish reason, latency, cost, final answer, observed and expected choice, normal-parse flag, correctness, and a projected tool trace. Each tool event contains its name, structured arguments, error flag, latency, and result hash. The release does not require raw provider credentials or hidden service state.

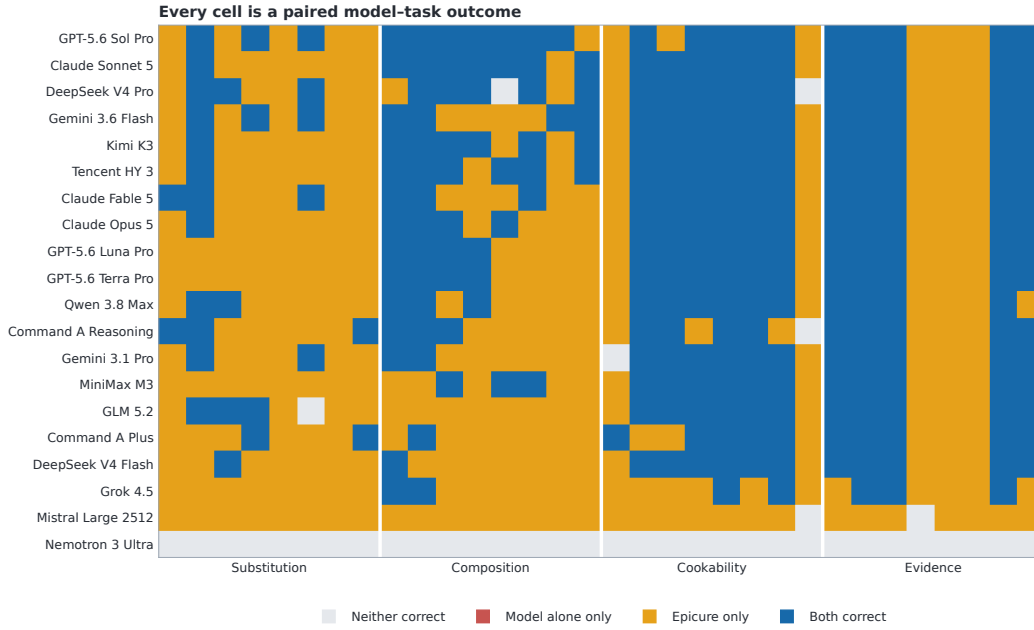


Figure 5: The complete 20×32 paired outcome matrix, ordered by public rank and grouped by task family. This is the most granular view of both benchmark difficulty and tool uplift.

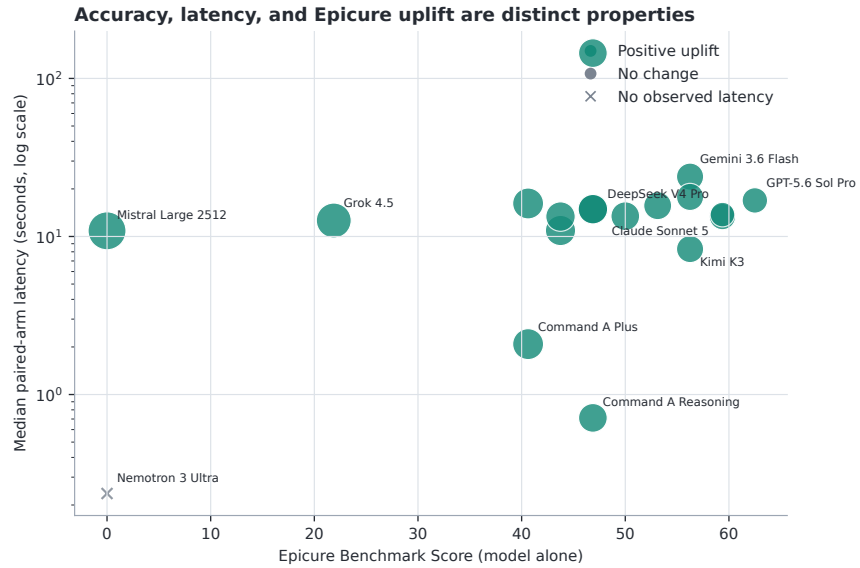


Figure 6: Score, latency, and tool uplift. Latency uses a log scale. Point size increases with the magnitude of uplift; colour indicates its direction. An $\times$ marks a route with no observed latency.

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 735 writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor,
 736 or originality of the research, declaration is not required.
 737 Answer: [Yes]
 738 Justification: The evaluated LLM endpoints are the experimental subjects. Section 3 and Ap-
 739 pendix C describe their paired role, tool exposure, routing, and failure handling.
 740 Guidelines:
 741 • The answer [N/A] means that the core method development in this research does not involve
 742 LLMs as any important, original, or non-standard components.
 743 • Please refer to our LLM policy in the NeurIPS handbook for what should or should not be
 744 described.